

Differential operators :- map which maps a function to another function.

$D = \frac{d}{dx}$ maps a function to its derivative.

• $D(y) = y'$

• $D(x^2) = 2x$

• $D(\cos 2z) = -2\sin 2z$

$D = \frac{d}{dz}$

• $D^2 = \frac{d^2}{dx^2}$

• $D^2 + aD + b$

Diff operators of second order

$$(D^2 + aD + b)y = y'' + ay' + by$$

• D is a linear operator ??

$$\begin{aligned} D(y_1 + y_2) &= Dy_1 + Dy_2 \\ D(ay) &= aD(y) \end{aligned} \quad \left. \vphantom{\begin{aligned} D(y_1 + y_2) &= Dy_1 + Dy_2 \\ D(ay) &= aD(y) \end{aligned}} \right\} \rightarrow D \text{ is a linear operator.}$$

• Non homogeneous linear diff Eqn.

$$y'' + p(x)y' + q(x)y = r(x) \quad \text{--- (1)}$$

Corresponding Homogeneous equation

$$y'' + p(x)y' + q(x)y = 0 \quad \text{--- (2)}$$

Thm :- (A) The difference of two solutions of (1) is a solution of Eqn (2).

and hence (A)

Proof ☺

is a solution of Eqn ②.

Proof of (A):- Let $y_1(x)$ and $y_2(x)$ are two solutions of ①,

$$y_1'' + p(x)y_1' + q(x)y_1 = r(x) \quad \text{--- (3)}$$

$$y_2'' + p(x)y_2' + q(x)y_2 = r(x) \quad \text{--- (4)}$$

$$\textcircled{3} - \textcircled{4} \quad y_1'' - y_2'' + p(x)(y_1' - y_2') + q(x)(y_1 - y_2) = 0$$

$$(y_1 - y_2)'' + p(x)(y_1 - y_2)' + q(x)(y_1 - y_2) = 0$$

$\Rightarrow y_1 - y_2$ is solution Eqn ②

⑥ The sum of one solution of Eqn ① and one solution of Eqn ② is a solution of ①

• y_1 is solution of ①

$$y_1'' + p(x)y_1' + q(x)y_1 = r(x) \quad \text{--- (5)}$$

• y_2 is solution of ②

$$y_2'' + p(x)y_2' + q(x)y_2 = 0 \quad \text{--- (6)}$$

$$\textcircled{5} + \textcircled{6} \quad (y_1 + y_2)'' + p(x)(y_1 + y_2)' + q(x)(y_1 + y_2) = r(x)$$

$\Rightarrow y_1 + y_2$ is solution of ①.

• How to find solution of Non-homogeneous diff Eqn.

$$y'' + p(x)y' + q(x)y = r(x)$$

Complete solution is

$$\left[\begin{array}{l} \text{solution of corresponding} \\ \text{Homogeneous diff Eqn} \\ \text{--- n. of constants} \end{array} \right] + \left[\begin{array}{l} \text{solution of} \\ \text{Non-Hom} \\ \text{diff Eqn} \\ \text{No. ... +} \end{array} \right]$$

Homogeneous will have contains No. of constants = order of diff Eqn
 No constant

① Solve $(D^2+4)y = x^2$

$$\frac{d^2y}{dx^2} + 4y = x^2$$

Corresponding Homogeneous Eqn :-

Ch Eqn
Solution

$$\frac{d^2y}{dx^2} + 4y = 0$$

$$r^2 + 4 = 0 \Rightarrow r = \pm 2i$$

$$y = C_1 \cos 2x + C_2 \sin 2x$$

• To solve Non-homogeneous linear diff Eqn with constant coefficients.

* Method of undetermined coefficients.

guess solution is $y = ax^2 + bx + c$

$$y' = 2ax + b$$

$$y'' = 2a$$

$$x^2 = y'' + 4y = 2a + 4ax^2 + 4bx + 4c$$

$$\Rightarrow 4ax^2 + 4bx + (2a + 4c) = x^2$$

$$4a = 1, \quad a = \frac{1}{4}$$

$$4b = 0, \quad b = 0$$

$$2a + 4c = 0$$

$$\frac{1}{2} + 4c = 0 \Rightarrow c = -\frac{1}{8}$$

$$y = \frac{1}{4}x^2 - \frac{1}{8}$$

Complete solution

$$C_1 \cos 2x + C_2 \sin 2x + \frac{1}{4}x^2 - \frac{1}{8}$$

Complete solution

$$y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{9}x^2 - \frac{1}{8}$$

corresponding homogeneous eqn

$$y'' + 4y = 0$$

$$y = C_1 \cos 2x + C_2 \sin 2x$$

• trial solution $y = Ae^{3x}$

$$y' = 3Ae^{3x}$$

$$y'' = 9Ae^{3x}$$

$$e^{3x} = 9Ae^{3x} + 4Ae^{3x} \Rightarrow 13Ae^{3x} = e^{3x}$$

$$\Rightarrow 13A = 1 \Rightarrow A = \frac{1}{13}$$

complete solution $y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{13}e^{3x}$.

• $y'' + y' - 2y = \sin x$

Corresponding homo eqn

ch eqn

solution

$$y'' + y' - 2y = 0$$

$$r^2 + r - 2 = 0 \quad r = -2, 1$$

$$y = C_1 e^{-2x} + C_2 e^x$$

Non-Homo

total solution $y = A \sin x + B \cos x$

$$y' = A \cos x - B \sin x$$

$$y'' = -A \sin x - B \cos x$$

$$\sin x = -A \sin x - B \cos x + A \cos x - B \sin x - 2A \sin x - 2B \cos x$$

$$\sin x = (-3A - B) \sin x + (-3B + A) \cos x$$

$$-3A - B = 1$$

$$A = -\frac{3}{10}$$

$$-3B + A = 0$$

$$B = -\frac{1}{10}$$

Complete solution

$$y = C_1 e^{-2x} + C_2 e^x - \frac{3}{10} \sin x - \frac{1}{10} \cos x$$

$r(x)$	total solution
polynomial in x of degree n	$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$
$k e^{ax}$	$A e^{ax}$ (A is constant to find)
$k \sin(ax)$ OR $k \cos(ax)$	$A \sin(ax) + B \cos(ax)$

• Solve $y'' + 2y' + y = x - e^x$

Homog Eqn

$$y'' + 2y' + y = 0$$

Ch Eqn

$$r^2 + 2r + 1 = 0$$

$$r = -1, -1$$

Solution

$$y = C_1 e^{-x} + C_2 x e^{-x}$$

Trial solution

$$y = Ax + B + Ce^x$$

$$y' = A + Ce^x, \quad y'' = Ce^x$$

$$x - e^x = Ce^x + 2A + 2Ce^x + Ax + B + Ce^x$$

$$x - e^x = 4Ce^x + Ax + (2A + B)$$

$$4C = -1 \Rightarrow C = -\frac{1}{4}$$

$$A = 1, \quad 2A + B = 0$$

$$B = -2$$

Complete solution

$$y = C_1 e^{-x} + C_2 x e^{-x} + x - 2 - \frac{1}{4} e^x$$

⑤ $y'' + 2y' + 4y = x \cos 3x$ $\xrightarrow{(Ax+B)}$

$\xrightarrow{\text{Homo Eqn}}$ $y'' + 2y' + 4y = 0$ $\xrightarrow{(C \cos 3x + D \sin 3x)}$
 $\xrightarrow{\text{Ch Eqn}}$ $\lambda^2 + 2\lambda + 4 = 0$ $\lambda = -1 \pm \sqrt{3}i$

$y_{\text{hom}} = e^{-x} (C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x)$

Non Homo Eqn total solution:-

$y = (Ax+B)(C \cos 3x + D \sin 3x)$

$$y = ACx \cos 3x + BC \cos 3x + ADx \sin 3x + BD \sin 3x$$

$$y = (ACx + BC) \cos 3x + (ADx + BD) \sin 3x$$

$$y = (K_1x + K_2) \cos 3x + (K_3x + K_4) \sin 3x$$

$$y' = K_1 \cos 3x + (K_1x + K_2)(-3 \sin 3x) + K_3 \sin 3x + (K_3x + K_4)(3 \cos 3x)$$

$$y' = (K_1 + 3K_4) \cos 3x + (-3K_2 + K_3) \sin 3x + 3K_3x \cos 3x - 3K_1x \sin 3x$$

$$y'' = (K_1 + 3K_4)(-3 \sin 3x) + (-3K_2 + K_3)(3 \cos 3x) + 3K_3(\cos 3x - 3x \sin 3x) - 3K_1(\sin 3x + 3x \cos 3x)$$

$$y'' = (-6K_1 - 9K_4) \sin 3x + (-9K_2 + 6K_3) \cos 3x - 9K_3x \sin 3x - 9K_1x \cos 3x$$

$$y'' + 2y' = (-9K_3 - 6K_1)x \sin 3x + (-9K_1 + 6K_3)x \cos 3x$$

$$y'' + 2y' = (-9k_3 - 6k_1)x \sin 3x + (-9k_1 + 6k_3)x \cos 3x \\ + (-6k_1 - 6k_2 + 2k_3 - 9k_4) \sin 3x \\ + (-9k_2 + 6k_3 + 2k_1 + 6k_4) \cos 3x$$

$$y'' + 2y' + 4y = (-5k_3 - 6k_1)x \sin 3x + (-5k_1 + 6k_3)x \cos 3x \\ + (-6k_1 - 6k_2 + 2k_3 - 5k_4) \sin 3x \\ + (-5k_2 + 6k_3 + 2k_1 + 6k_4) \cos 3x$$

||

$$x \cos 3x$$

k_1 & k_3

k_2 & k_4

$$\left\{ \begin{array}{l} -5k_3 - 6k_1 = 0 \\ -5k_1 + 6k_3 = 1 \end{array} \right.$$

$$\left\{ \begin{array}{l} -6k_1 - 6k_2 + 2k_3 - 5k_4 = 0 \\ -5k_2 + 6k_3 + 2k_1 + 6k_4 = 0 \end{array} \right.$$

